Pradip Sapkota

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Education

Minnesota State University, Mankato — Mankato, MN
B.S. Computer Engineering, GPA: 3.7/4.0 — *Expected Dec. 2026*

Technical Skills

Programming: C, C++, Python, MATLAB, SystemVerilog, VHDL, SQL, Ladder Logic
Hardware & Embedded: FPGA, Microcontrollers, Raspberry Pi, Sensors, Digital Logic, PLC, Modbus TCP
Engineering Tools: KiCad, Autodesk EAGLE, Oscilloscope, Spectrum Analyzer, AutoCAD, Autodesk Revit, SolidWorks, ModelSim
Testing & Systems: Hardware Debugging, System Validation, Linux, Git/GitHub, AWS Lambda, DynamoDB

Experience

Minnesota State University, Mankato — Jan. 2024 – Present
IT Solutions Consultant — *Mankato, MN*

- Resolved **50+ technical incidents** involving computing hardware, software, and connectivity using systematic fault isolation.
- Diagnosed device, configuration, and authentication issues through root-cause analysis to identify corrective steps.
- Supported device setup, validation, and documentation to restore operation for university technology users.

Ruppell Inc. — Jul. 2025 – Aug. 2026
Engineering Associate – Electronics and Software — *Mankato, MN*

- Collaborated with PCB design engineers on approximately **1–2 product projects**, supporting component integration and design discussions.
- Assisted hardware test teams with functional testing and troubleshooting to identify issues for product debugging.
- Contributed frontend code and UI layouts for the company website and product dashboards to improve information presentation.

Minnesota State University, Mankato — Aug. 2023 – Jan. 2024
Undergraduate Research Assistant – FPGA & Embedded Systems — *Mankato, MN*

- Supported FPGA research using VHDL logic design and simulation to evaluate digital circuit functionality.
- Assisted with STM32 microcontroller research and hardware testing to investigate embedded system behavior.
- Recorded test observations and technical findings to support debugging and design verification.

Projects

Smart Door Access Control System | *KiCad, Autodesk EAGLE, Embedded Systems, Python* — Apr. 2025

- Designed and fabricated an access-control PCB using KiCad and Autodesk EAGLE to integrate locks, sensors, and authentication hardware.
- Implemented access-control logic with anti-spoofing and fail-safe behavior; performed reliability testing across hardware and software.

Interactive AI Voice Assistant | *KiCad, Autodesk EAGLE, Raspberry Pi, Python* — Dec. 2024

- Designed and fabricated a PCB using KiCad and Autodesk EAGLE for a Raspberry Pi voice assistant with audio and GPIO-controlled hardware.
- Integrated Python software, audio, GPIO control, and external AI services; debugged hardware–software interactions during system testing.

Leadership

Institute of Electrical and Electronics Engineers (IEEE) — Aug. 2023 – Aug. 2024
Events Lead — *Mankato, MN*

- Lead embedded-design competitions and technical workshops, coordinating schedules, equipment, logistics, and faculty/student teams.